How Much of Reported Quantization Degradation Is Measurement? Decomposing Accuracy Loss into Competence, Compliance, and Truncation

Anonymous ACL submission

Abstract

[**TODO:** Abstract — written in W5 (C7), after results freeze.]

1 Introduction

2 Related Work

3 Method

3.1 Experimental design

3.2 Quantization and calibration

3.3 The scoring ladder

3.4 Compliance, competence, and truncation

3.5 Serving configuration and determinism

4 Results

4.1 Truncation (H1)

[**TODO:** H1 — degradation vs generation budget; mean/median n_gen_tokens by precision with CIs, settling 2606.25519 vs 2504.04823.]

4.2 Recoverable failures (H2)

[**TODO:** H2 — gold-in-trace rate, permissive and conservative variants, always reported net of the incidental-match false-positive floor.]

4.3 Precision × scoring interaction (H3)

[**TODO:** H3 — GLMM interaction as an odds ratio and as a marginal effect in accuracy points with 95% CI, plus paired McNemar. A NULL IS PUBLISHABLE and must not read as failure.]

4.4 Decomposition (H4)

[**TODO:** H4 — THE CENTREPIECE TABLE. Design it first and let it drive what gets logged. Always-compliant stratum size as a first-class number; compliance-transition matrix; Manski bounds.]

4.5 Prompt format (H5) and the structural control (H6)

[**TODO:** H5 zero-shot arm; H6 ARC-Challenge under log-likelihood vs generative scoring on the same 500 items — the gap between modes IS the extraction cost.]

4.6 Statistical resolution of published deltas (H7)

[**TODO:** H7 — scrupulously neutral: "many papers report aggregate deltas without a paired test", NEVER "the deltas are noise".]

4.7 Bitwise reproducibility of quantized serving (H8)

[**TODO:** H8 — per-format bitwise disagreement under a batch-invariant engine. Self-contained result nobody has reported; may become a headline.]

5 Discussion

6 Conclusion

Limitations

Ethics Statement

Acknowledgments

References

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A Frozen configuration

B Annotation rubric

C Pre-registration